Monsoon AI

TEAM NAME: Monsoon

TEAM MEMBERS:

No.	Full Name	Role
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1. PROJECT OVERVIEW

1.1 Project Title: Monsoon AI

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1.2 Key challenge areas: Urban Air Quality and Climate Resilience

1.3 Solution Category: Mobile App/Web App

1.4 Project Summary

When meteorological agencies issue severe weather alerts, millions of vulnerable citizens worldwide receive the same generic regional warnings that fail to translate into survivable household decisions, a failure mode that cost over 6,300 lives during Typhoon Haiyan and 200+ during the 2021 German floods. MonsoonAI is a dual-interface climate resilience platform that fuses official meteorological data, satellite aerosol indices, AI-driven river discharge forecasts, and LiDAR flood topography with custom household profiles to deliver hyper-localized, plain-language action plans via two-way SMS and a web app. A deterministic risk engine paired with LLM-only output translation (GPT-5 mini) eliminates hallucination in life-safety calculations, while a Predictive Triage Dashboard enables local governments to pre-deploy rescue resources 24 hours before landfall. Built on open data feeds and an ASEAN-native swappable architecture, MonsoonAI deploys initially across the Philippines and Vietnam and is positioned to scale globally under the UN Early Warnings for All initiative.

2. CONTEXT AND PROBLEM STATEMENT

During severe calamities such as typhoons, floods, and transboundary environmental hazards, official early warning systems worldwide rely on generic, region-wide alerts that fail to translate into survivable decisions for high-risk populations. According to the World Meteorological Organization, one in three people globally still lack access to effective early warning systems. The consequences are fatal across both developing and developed nations: in 2013, PAGASA correctly issued a storm-surge warning 18 hours before Typhoon Haiyan, but the terminology was not understood at household level, costing over 6,300 lives; in coastal Vietnamese provinces, power-grid dependent loudspeakers create communication blind spots during typhoon-induced outages; the 2015 Indonesian peatland fires sent transboundary haze across Southeast Asia, linked to an estimated 100,000 premature deaths; and in July 2021, Germany's EFAS issued accurate flood warnings days in advance, yet what followed was what its own designer called "a monumental failure," with over 200 deaths. These failures demand urgent attention as the IPCC's Sixth Assessment Report projects continued increases in tropical cyclone intensity and compound flood events. Primary stakeholders are high-risk citizens, the elderly, persons with disabilities, children, and households in structurally vulnerable housing. Secondary stakeholders are LGUs, the Philippines' NDRRMC, Vietnam's VNDMA, and equivalent disaster management authorities worldwide.

3. EXISTING SOLUTIONS AND GAP ANALYSIS

While existing systems have built strong individual components, vulnerable-persons registries, community volunteer networks, and two-way SMS pilots, no country has integrated all five operational layers into a single platform, with even the most advanced systems failing at the last mile, from Typhoon Haiyan's 6,300 lives lost to terminology misunderstood at household level, to flood warnings their own designer called "a monumental failure" after 180+ deaths. MonsoonAI fills this global integration gap by consuming the same open data these systems already produce and adding the household-level, plain-language, two-way, anticipatory layer that architectural, institutional, and operational barriers have prevented any existing solution from delivering.

✓ Fully present | ⚠ Partially present | ✗ Absent

Feature	MonsoonAI	Philippines	Vietnam	Japan	USA	Germany / EU	Bangladesh
1. Household-level Shelter vs. Evacuate decision	✓	✗	✗	⚠	⚠	✗	✗
2. Air Quality & Heat Advisory with transboundary haze	✓	⚠	✗	⚠	⚠	✗	✗
3. Predictive pre-storm Rescue Priority Manifest (B2G)	✓	✗	✗	⚠	⚠	✗	✗
4. Automated nearest-neighbor warden dispatch	✓	✗	✗	✗	✗	✗	⚠
5. Two-way SMS query for live status	✓	✗	⚠	✗	✗	✗	✗

4. PROPOSED SOLUTION AND CORE FEATURES

Solution Overview: MonsoonAI is a 365-day climate resilience platform that fuses official meteorological data, satellite imagery, and household profiles to deliver hyper-localized, plain-language alerts via SMS and web app, while powering a Predictive Triage Dashboard that enables LGUs to pre-deploy rescue resources before extreme weather strikes.

Feature 1 - Shelter vs. Evacuate Engine: A deterministic engine cross-validates flood signals and calculates personalized evacuation route times to the nearest shelter. GPT-5 mini translates results into plain language, performing zero calculations to eliminate hallucination risk.

Feature 2 - Dual-Layer Air Quality & Heat Advisory: Satellite data detects local fire smoke and transboundary haze anomalies against a self-calibrating baseline, generating year-round targeted health advisories by user profile, warning asthmatics, alerting the elderly, and flagging dangerous heat indexes.

Feature 3 - Predictive LGU Triage Manifest (B2G): Twenty-four hours before landfall, the dashboard auto-generates a ranked Rescue Priority List scoring households by structural vulnerability and demographic risk, enabling LGUs to pre-deploy rescue teams via one-tap routing.

Feature 4 - Micro-Responder Dispatch Protocol: When a vulnerable user triggers a critical alert, a PostGIS nearest-neighbor query automatically pings the closest enrolled community warden for a visual check or evacuation assistance without straining central rescue boats.

Feature 5 - Two-Way SMS Query Bot: Citizens text keywords (STATUS, EVAC, FLOOD) to receive near-instant, localized plain-text responses via a strict keyword router, no NLP, no smartphone required, and citizens can register their household settings through the same bot.

5. INNOVATION & COMPETITIVE ADVANTAGE

Innovation - Anticipatory Triage Manifesting: Traditional disaster management relies on reactive rescue deployment after distress calls. MonsoonAI shifts this paradigm by operating 24 hours before landfall: the deterministic engine generates a dynamically scored Rescue Priority Manifest, mathematically ranking households by structural vulnerability and demographic risk, enabling LGUs to pre-deploy resources to the exact coordinates of highest need before the hazard strikes.

Competitive Advantage - Bridging the Digital and Linguistic Divide: While current advanced warning systems depend on smartphones and high-speed internet, MonsoonAI acknowledges that the most vulnerable individuals lack these tools. Alerts are deployed in clear, localized language via SMS that doesn't require a sophisticated phone, providing low-income households, the elderly, and remote communities with the same vital information available to residents in developed cities. With the ability to integrate any language and data source instantly, protection extends beyond ASEAN.

Feasibility & Development Plan: *Hackathon:* The full five-feature architecture runs live on Node.js, PostGIS, Twilio SMS, NASA FIRMS, and Open-Meteo, with GloFAS, TROPOMI, and Zalo OA mocked via pre-loaded scenarios on zero-cost infrastructure. *Phase 1 (Months 1–6):* Mocked integrations go live and pilot deployments begin in one Philippine barangay and one Vietnamese commune. *Phase 2 (Months 6–18):* The LGU Dashboard is packaged as SaaS through NDRRMC and VNDMA procurement channels, with architecture replication-tested for a third ASEAN country. *Phase 3 (Months 18–36):* MonsoonAI formally aligns with the UN Early Warnings for All initiative, expanding to SIDS and South Asian coastal nations. MonsoonAI is technically feasible through reliance on free, open-licensed REST

data sources and a zero-app-download HTML5 PWA; financially sustainable through near-zero infrastructure costs and a B2G SaaS model tied to existing LDRRM procurement; and regulatorily viable through consent-first onboarding mirroring existing Barangay Tanod registration.

6. TARGET GROUPS & POTENTIAL IMPACT

Target Audience - MonsoonAI targets vulnerable communities in disaster-prone regions: the elderly facing mobility and evacuation challenges; persons with disabilities requiring accessible alerts; children dependent on guardians; low-income families lacking resilient housing or reliable communications; flood-prone households facing rapid inundation; residents of weakly constructed homes; individuals reliant on basic SMS-capable mobile devices; and Local Government Units requiring real-time situational awareness for resource deployment. Secondary users include disaster response agencies and environmental authorities.

Potential Impact - MonsoonAI has the potential to reduce deaths and casualties from natural disasters by providing localized, easy-to-comprehend evacuation alerts. Its predictive triage model enables faster emergency response by assisting local governments with operational resource allocation before disasters strike. MonsoonAI also supports public health through air quality and heat advisories that reduce population exposure to unhealthy environmental conditions, while promoting inclusive, accessible, disaster-resilient communities through community-based response features.

7. DESCRIPTION OF TECHNOLOGIES APPLIED

A. Proposed Technologies - Node.js with Express handles backend logic and Twilio SMS inbound webhooks. PostgreSQL with PostGIS manages household geocoordinates and nearest-neighbor spatial queries on Supabase free-tier infrastructure. The deterministic risk engine performs all life-safety calculations; GPT-5 mini handles output translation only into Tagalog and Vietnamese. The citizen-facing interface is a Vanilla HTML5/JavaScript Progressive Web App; the LGU Dashboard is built with React and Leaflet.js.

B. System Architecture & Technical Approach - A cron-triggered ingestion layer polls Open-Meteo (ECMWF) precipitation, NASA FIRMS VIIRS fire hotspots, Sentinel-5P TROPOMI aerosol indices, and GloFAS river discharge on scheduled intervals. The deterministic risk engine cross-validates dual signals (rainfall against UP NOAH 25-year inundation polygons AND river discharge thresholds) before firing any alert. Schema-locked GPT-5 mini translation converts risk JSON into bounded urgency messages. Twilio (PH) and Zalo OA (VN) handle SMS delivery; React + WebSocket pushes real-time updates to the LGU dashboard.

C. Data and Infrastructure Requirements - All primary data sources are free and open: UP NOAH flood polygons (Open Database License), NASA FIRMS, Sentinel-5P TROPOMI via Google Earth Engine, Open-Meteo, GloFAS, OpenStreetMap (Nominatim, OSRM), and OpenTopoData SRTM elevation. Infrastructure runs on Supabase free tier for the hackathon demo, with auto-scaling AWS or GCP planned for typhoon-season production traffic. Semaphore SMS (₱0.50/SMS) handles Philippine outbound bulk delivery.

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